

Physiology of respiratory system

Respiration is the process by which the body takes in and utilizes oxygen (O_2) and gets rid of carbon dioxide (CO_2).

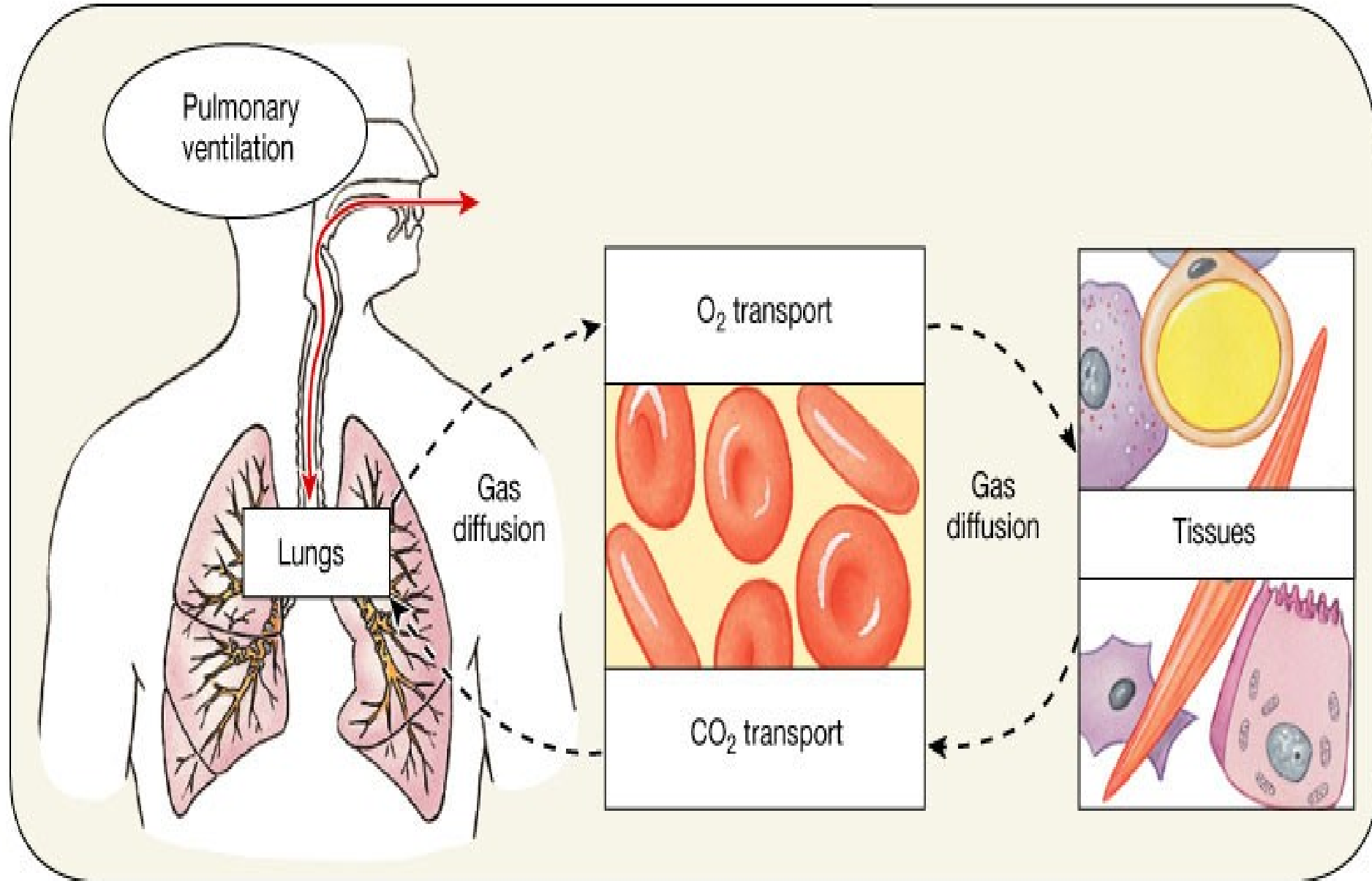
Respiratory System Functions

- **Gas exchange:** Oxygen enters blood and carbon dioxide leaves
- **Regulation of blood pH:** Altered by changing blood carbon dioxide levels
- **Voice production:** Movement of air past vocal folds makes sound and speech
- **Olfaction:** Smell occurs when airborne molecules drawn into nasal cavity
- **Protection:** Against microorganisms by preventing entry and removing them
- **Metabolism:** Synthesize and metabolize different compounds

Respiration can be divided into four major functional events

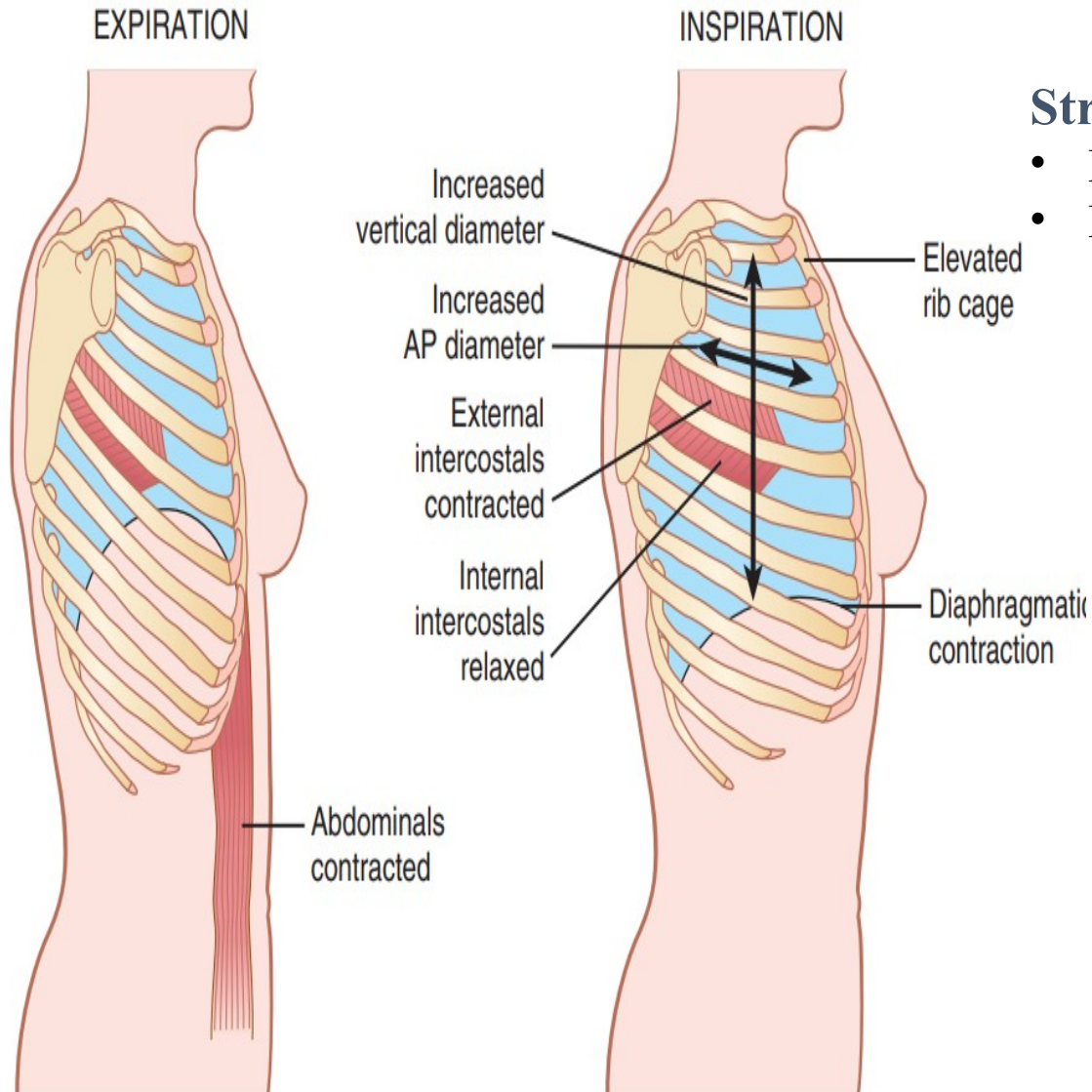
- **Ventilation:** Movement of air into and out of lungs
- **Gas exchange** between air in lungs and blood
- **Transport** of oxygen and carbon dioxide in the blood
- **Internal respiration:** Gas exchange between the blood and tissues

Key Steps in Respiration



Mechanics of ventilation

structures and Pressure



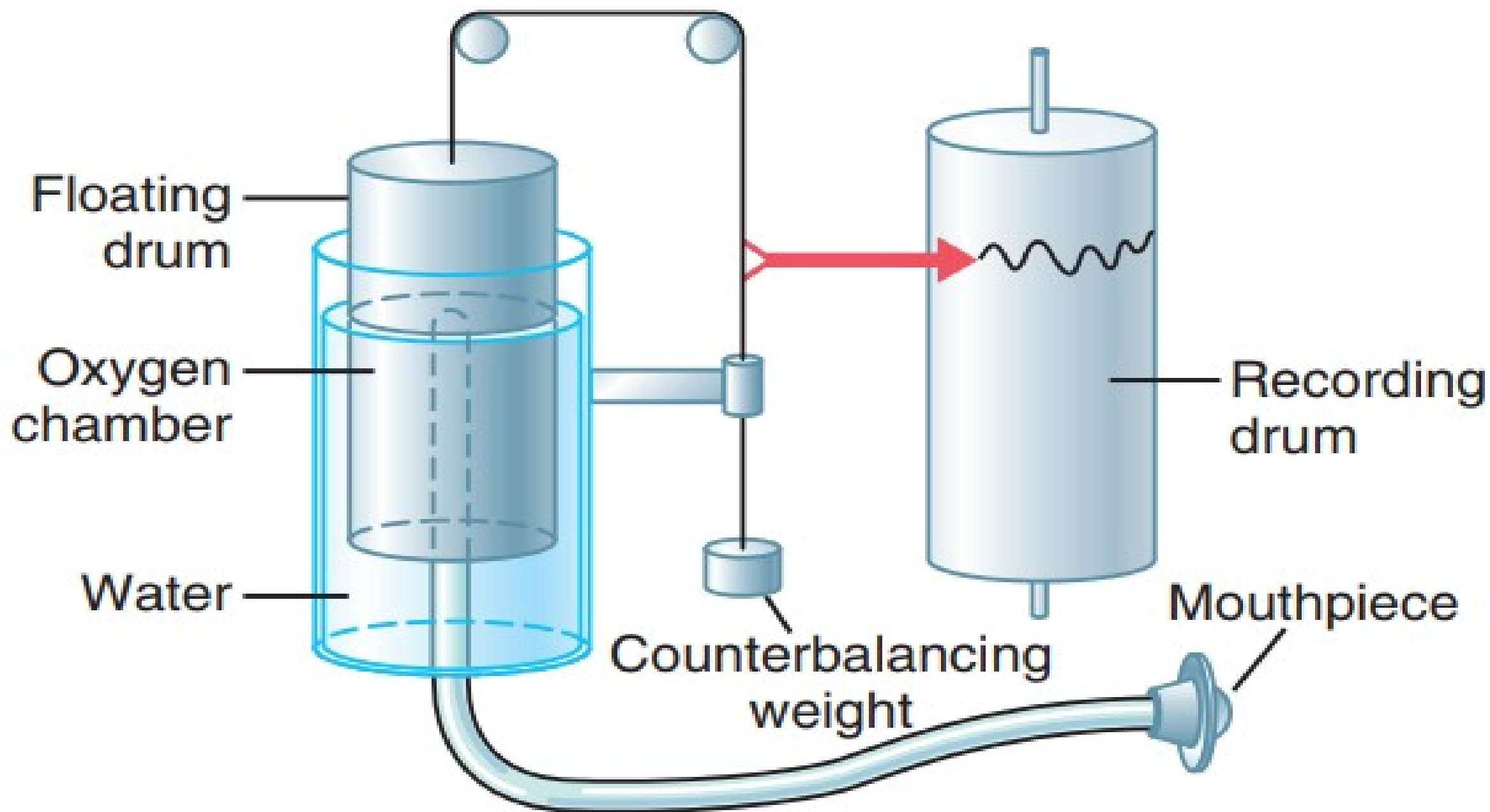
Structures:

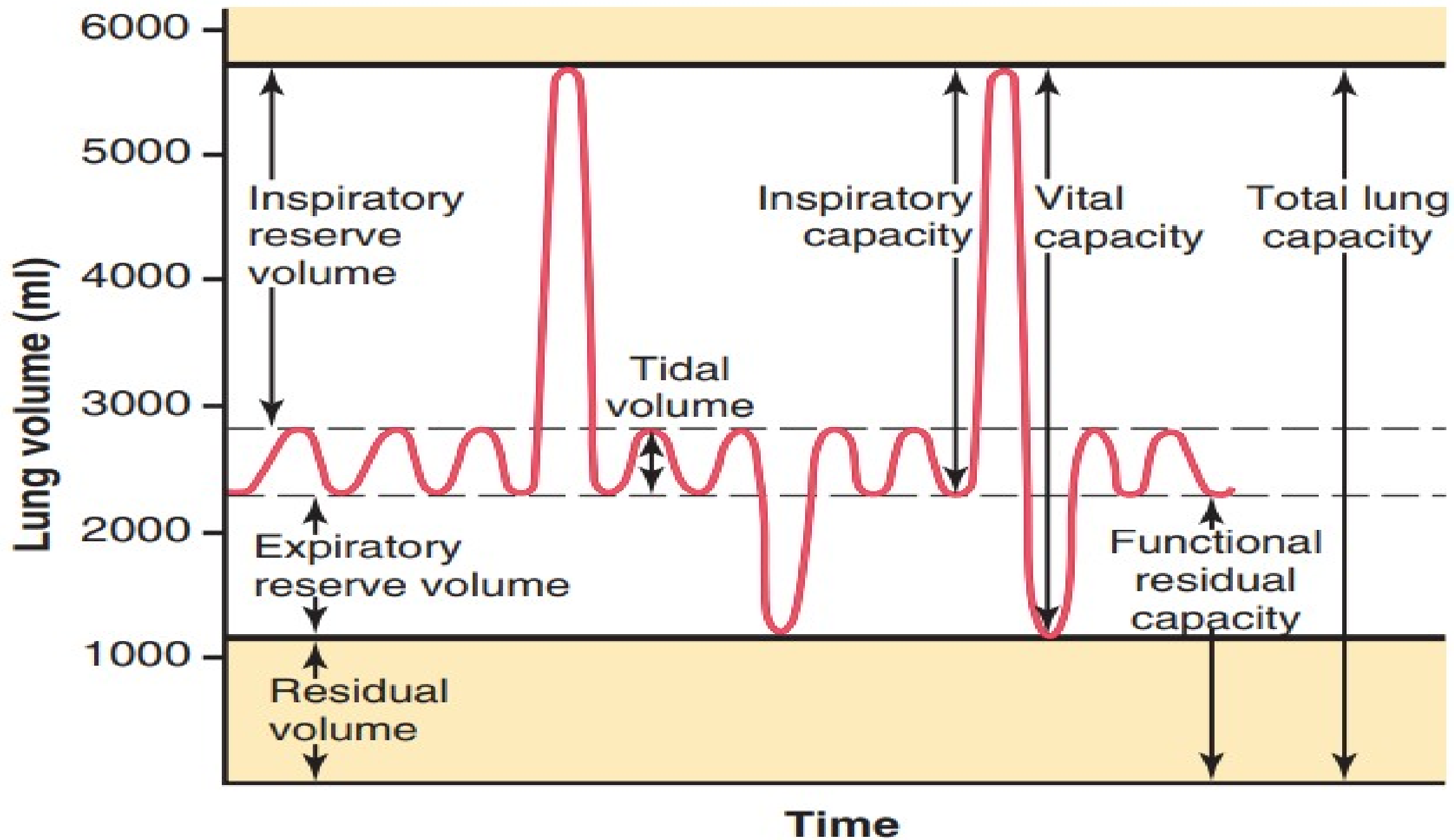
- Diaphragm
- Ribs and rib cages
 - ❖ muscles of inspiration_ external intercostal
 - ❖ muscles of expiration_ internal intercostal

Pressure:

- Intra pleural pressure: -5cm of water
- Intra alveolar Pressure: below 0cm of water
- Trans pulmonary Pressure: a measure of the elastic forces in the lungs

Compliance of the Lungs: The extent to which the lungs will expand for each unit increase in trans pulmonary pressure



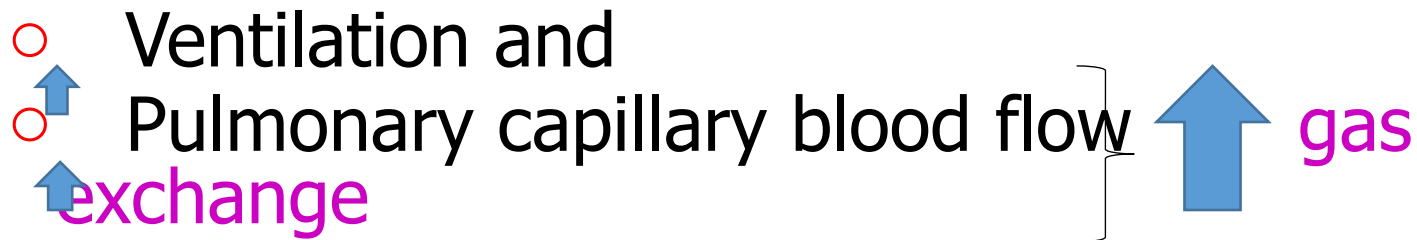


Physical Principles of Gas Exchange

❖ Diffusion of gases through the **respiratory membrane** depends on:

1. membrane's thickness,
2. diffusion coefficient of gas,
3. surface areas of membrane,
4. partial pressure of gases in alveoli & blood.

❖ Relationship between **ventilation** & pulmonary capillary **flow**



O₂ & CO₂ diffusion gradients

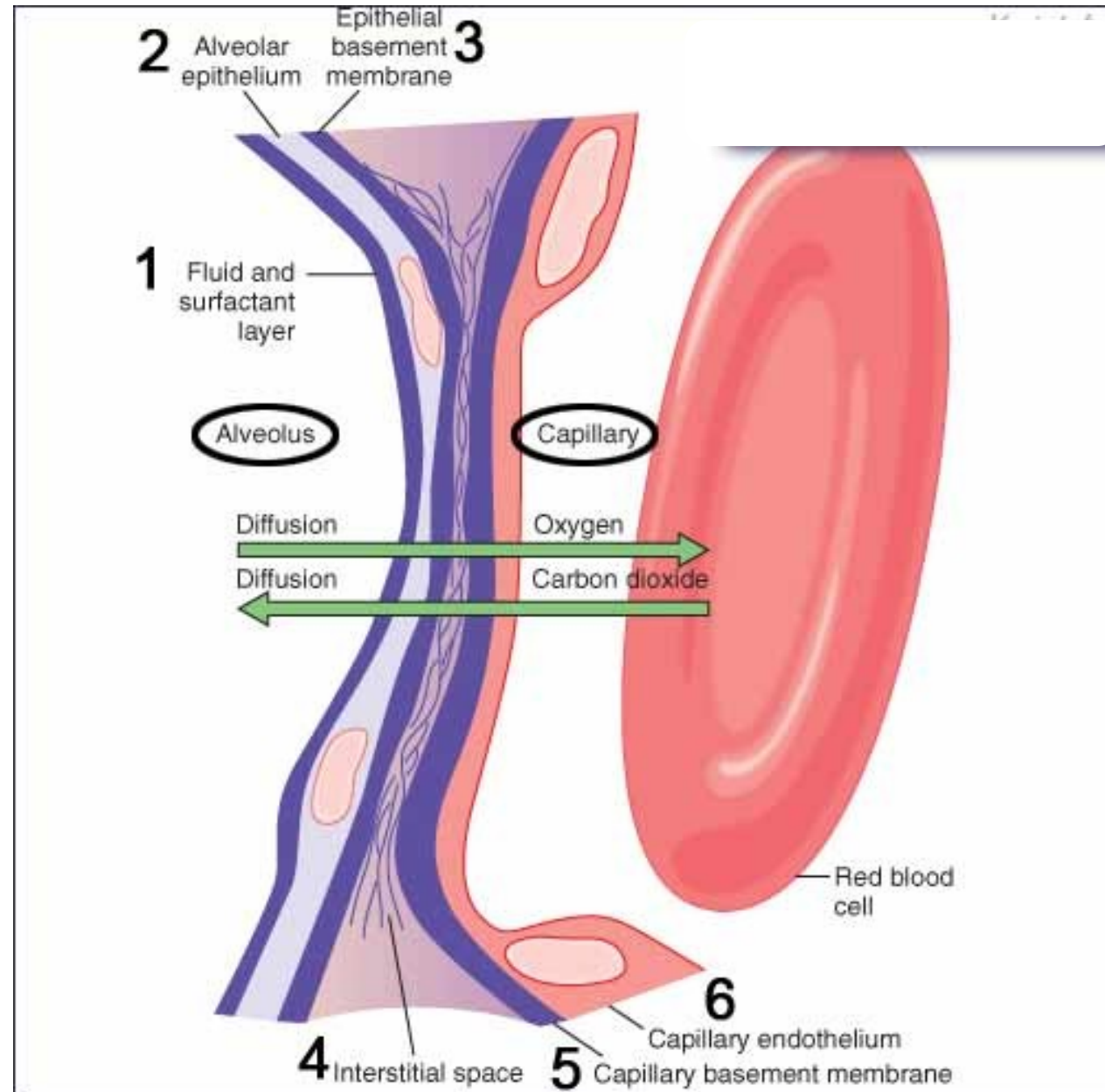
O₂

- Moves **from alveoli** into blood. **Blood** is almost completely **saturated** with **O₂** when it leaves the **capillary**.
- P_{O₂} in blood ↓ because of **mixing** with deoxygenated blood.
- **O₂** moves from tissue capillaries into the tissues.

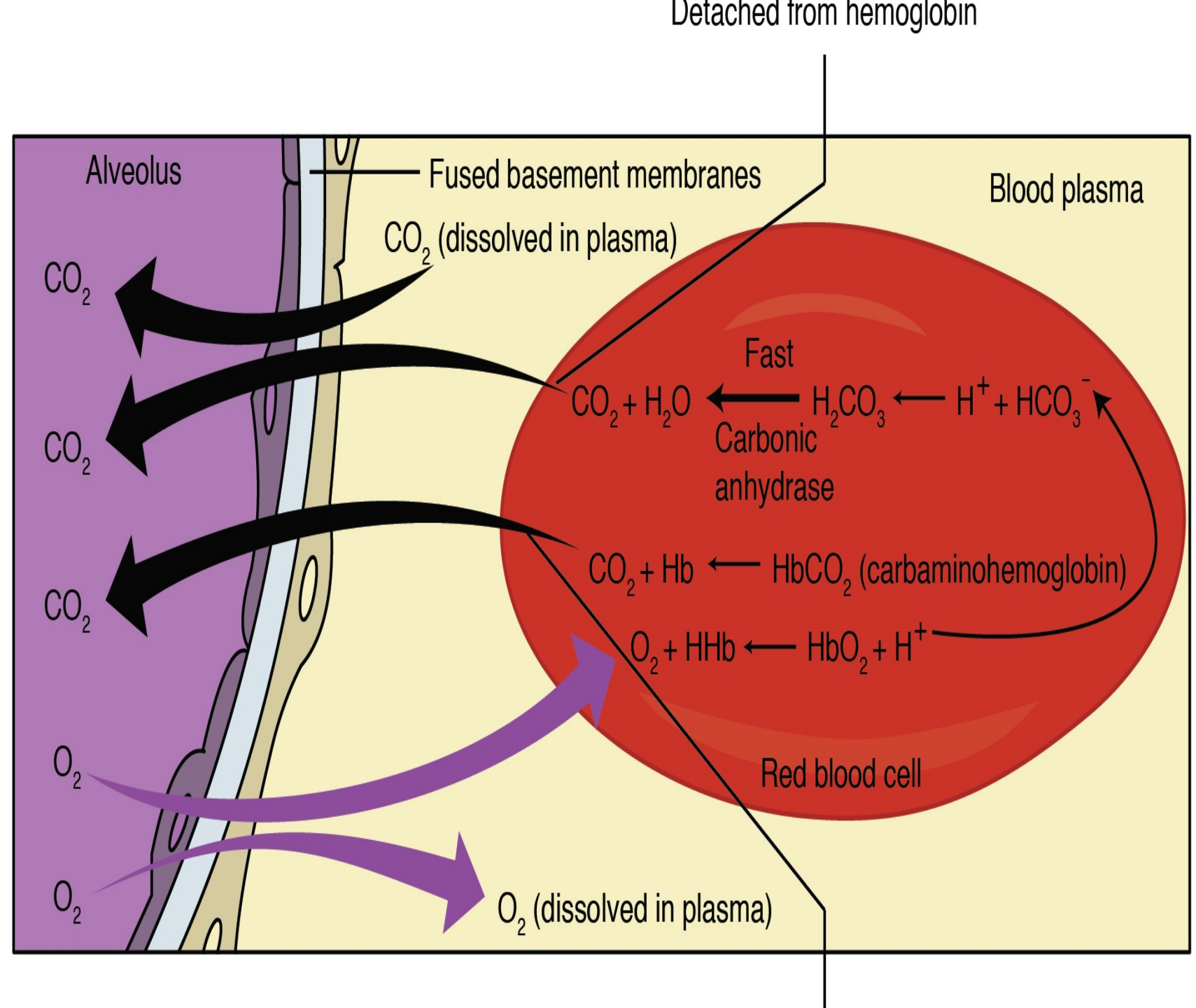
CO₂

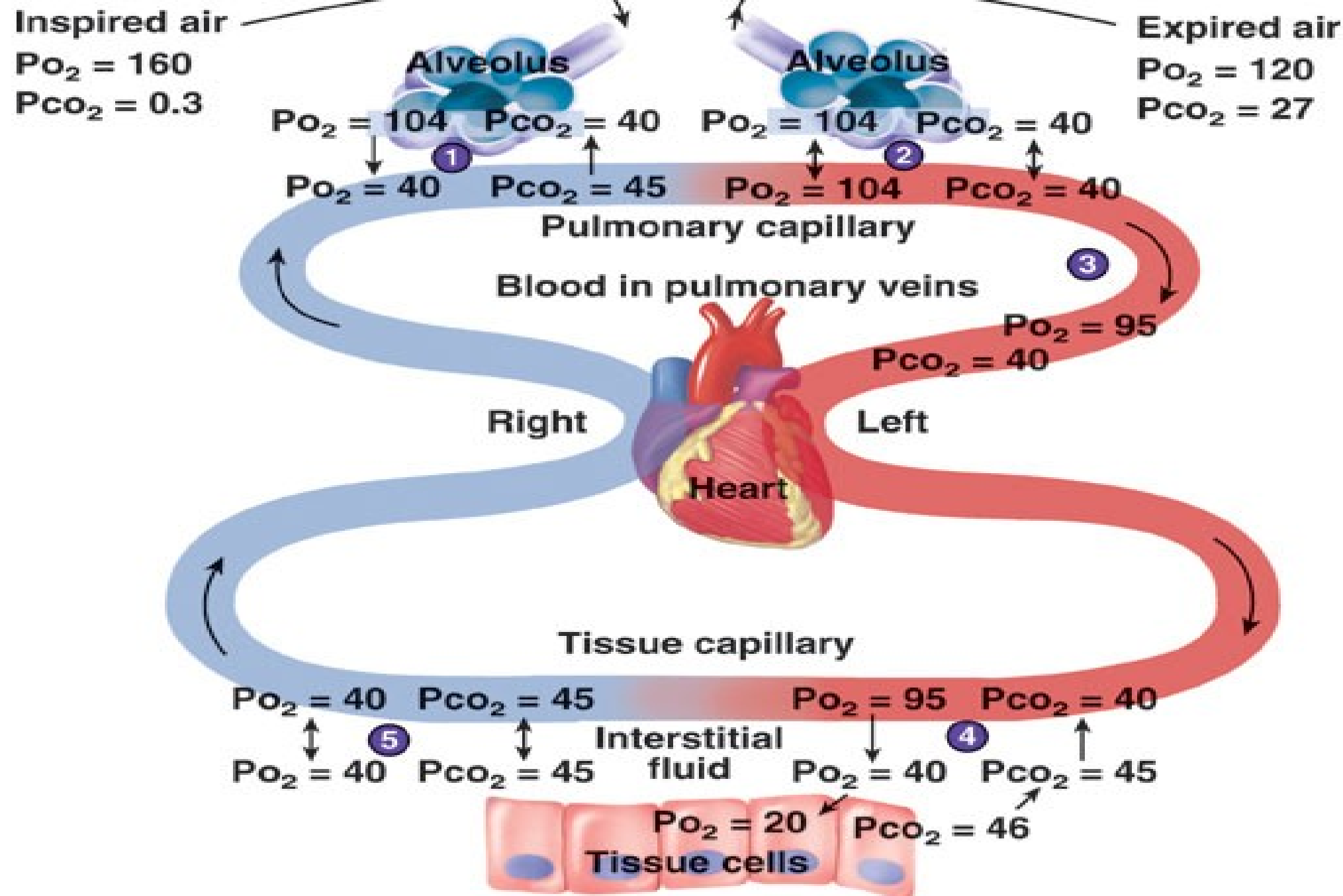
- Moves **from tissues** into tissue capillaries.
- Moves from pulmonary capillaries into the alveoli.

O_2 & CO_2 diffusion gradients...

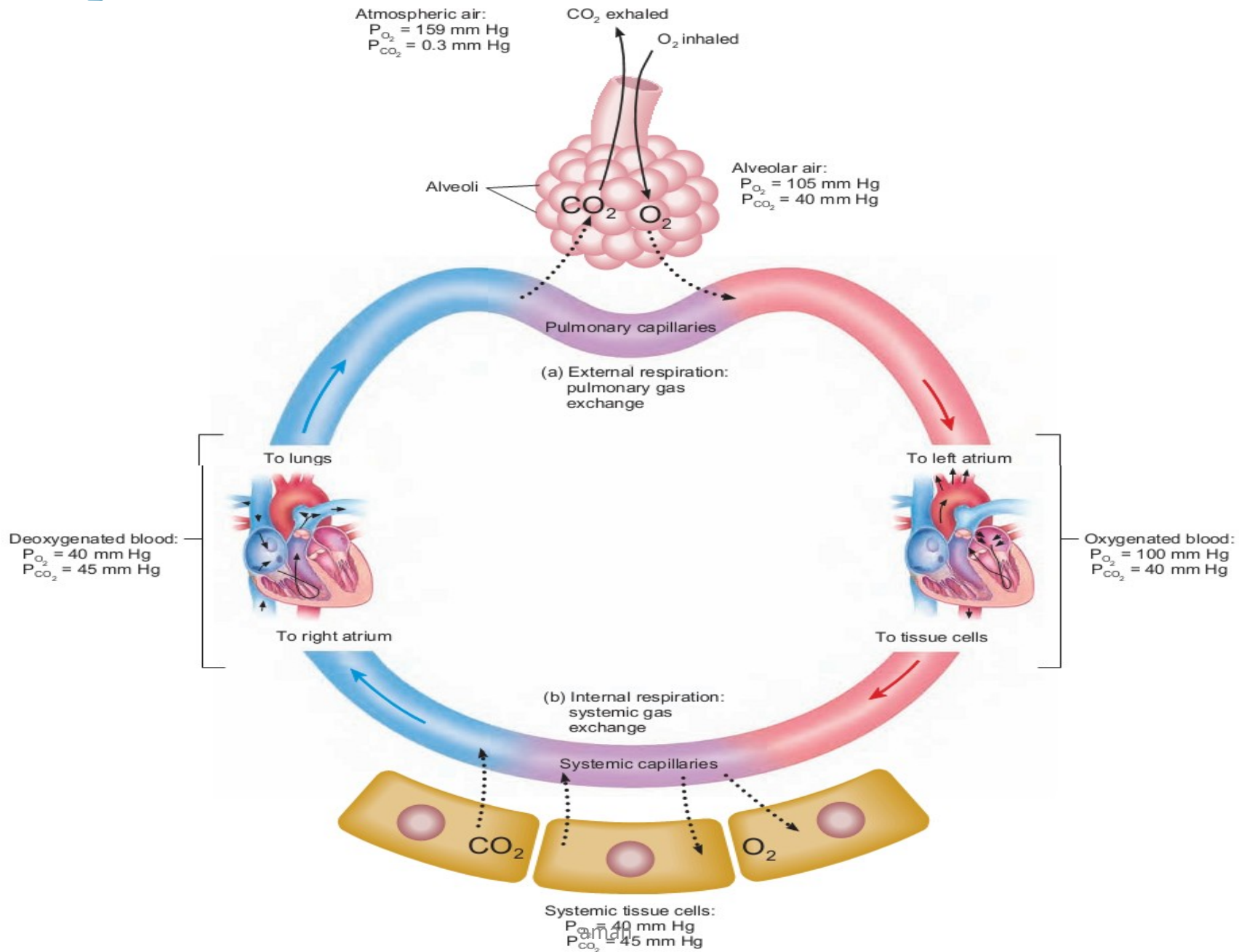


Blood gas transport...





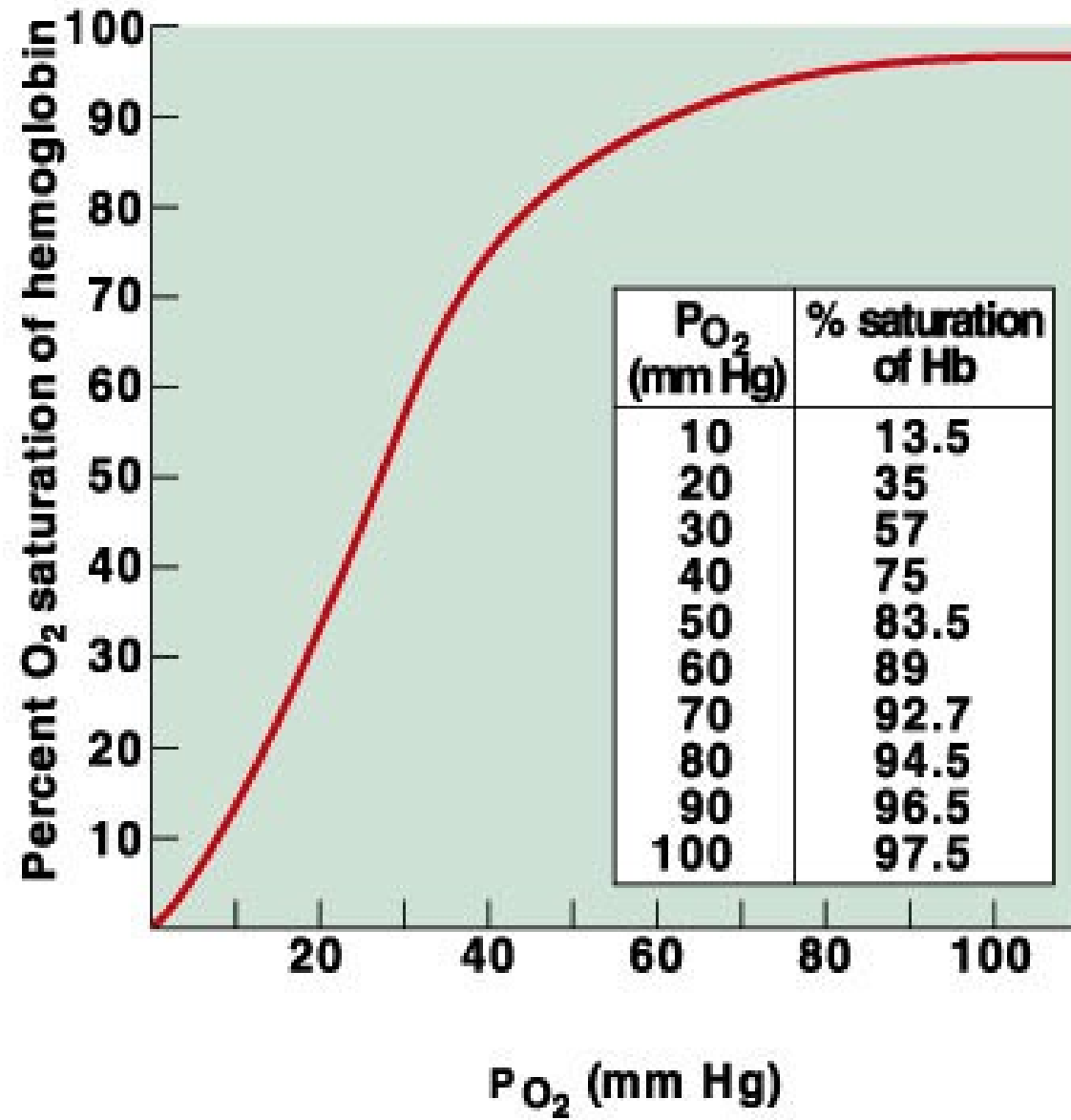
Each gas in a mixture of gases diffuses from an area of higher partial pressure of that gas to an area of lower partial pressure of that gas.



Oxygen Transport

- **Oxygen** is transported by:
 1. **Hemoglobin** (97%) and
 2. Dissolved form in **plasma** (3%)
- **O₂-hemoglobin dissociation curve** shows that **hemoglobin is almost completely saturated** when **P_{O₂}** is **80 mmHg** or above.
- At **lower partial pressures**, hemoglobin releases **O₂**.
- The substance **2,3-bisphosphoglycerate** ↑ the ability of **hemoglobin** to **release O₂**.

**Oxygen
Hemoglobin
dissociation
curve
at
rest**

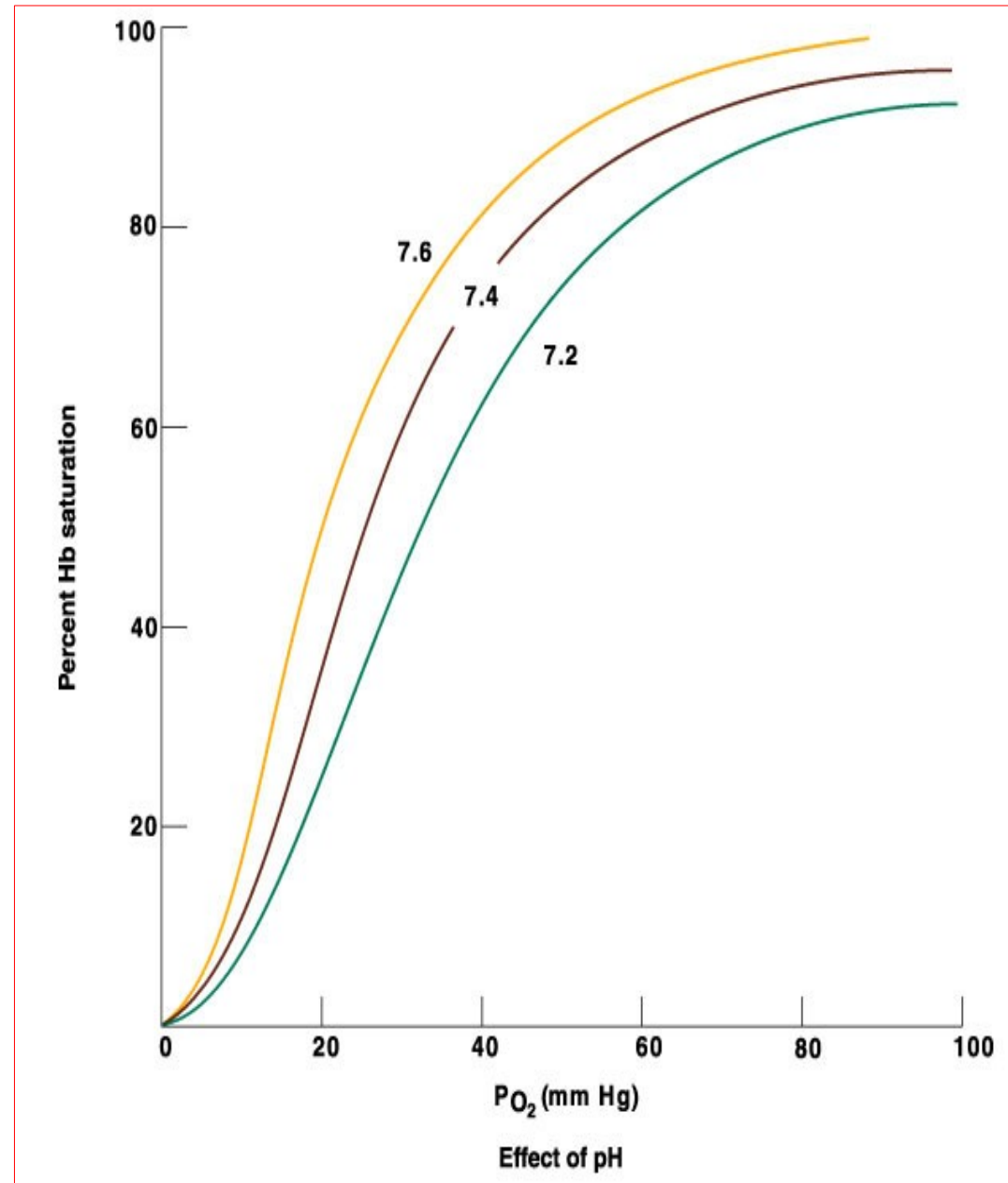


Oxygen Transport...

- A shift of the curve to the **left** because of
 - an ↑ in **PH**,
 - a ↓ in **CO₂** or a ↓ in **temperature**
 - results in an ↑ in the ability of hemoglobin to hold O₂.
- A shift of the curve to the **right** because of
 - a ↓ in **PH**,
 - an ↑ in **CO₂** or an ↑ in **temperature**
 - results in a ↓ in the ability of hemoglobin to hold O₂.

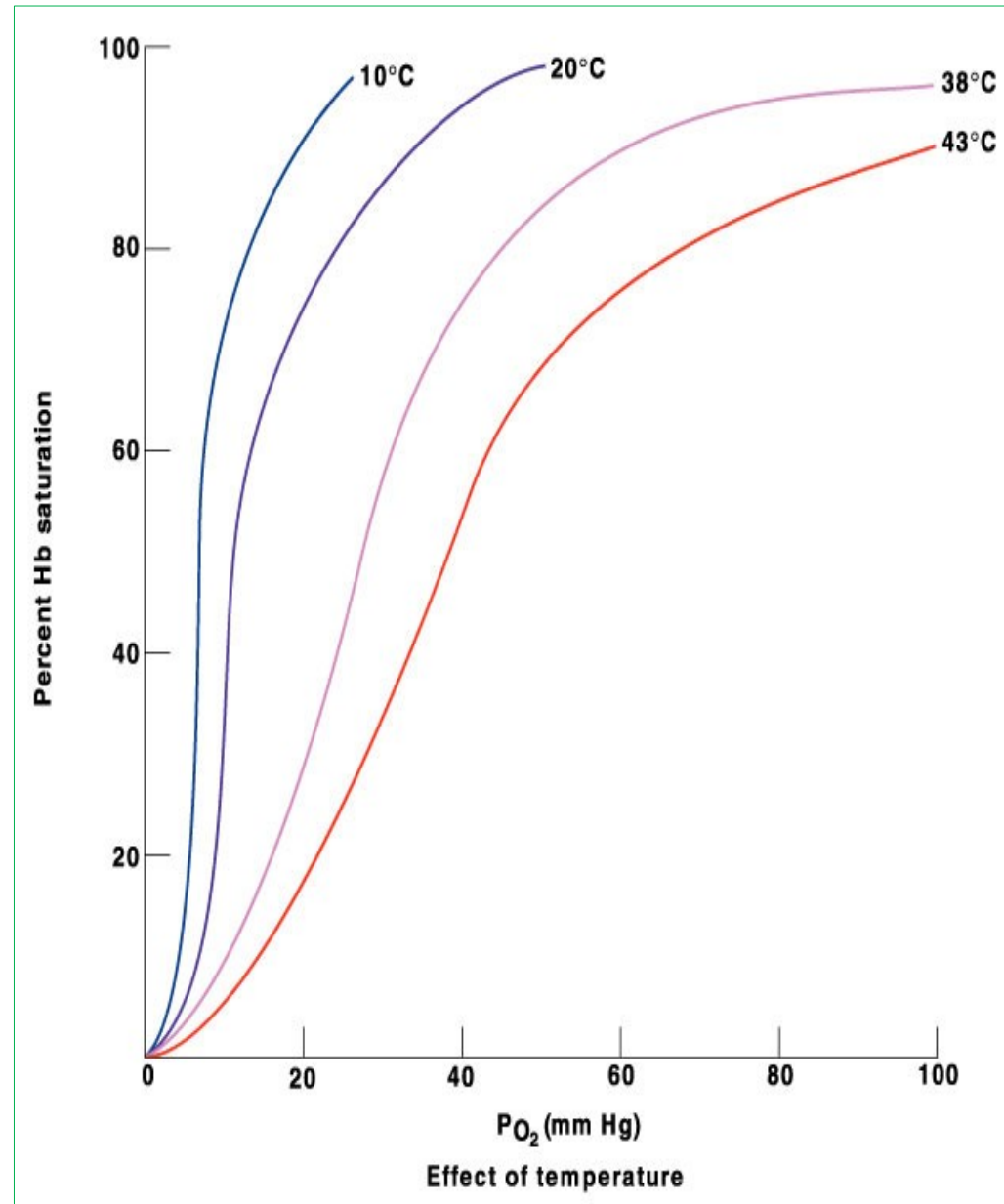
Bohr effect

- An $\uparrow$ in CO_2 in the blood causes O_2 to be displaced from the hemoglobin.
- Which is an **important factor in $\uparrow$ O_2 transport.**



Temperature effects:

- An $\uparrow$ in temperature results in a $\downarrow$ in the ability of hemoglobin to hold O_2 .

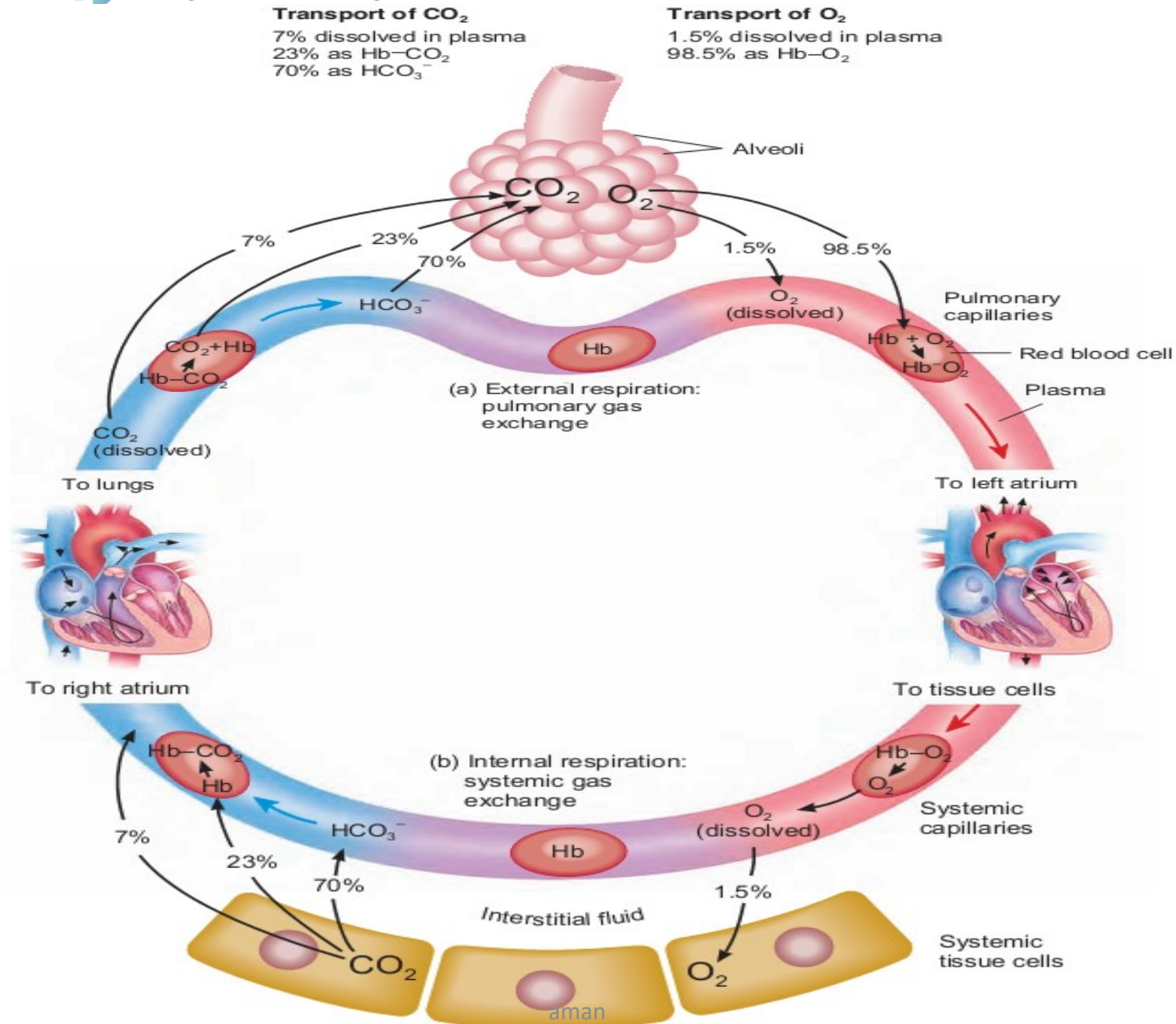


Transport of Carbon Dioxide

- **Carbon dioxide** is transported in the form of;
 1. **Bicarbonate** -70%
 2. **Carbaminohemoglobin** - 23%
 3. Dissolved in **plasma** - 7%
- **Haldane effect**
 - Hemoglobin that has released O_2 binds more readily to CO_2 than hemoglobin that has O_2 bound to it.



Most O_2 is transported by hemoglobin as oxyhemoglobin within red blood cells; most CO_2 is transported in blood plasma as bicarbonate ions.



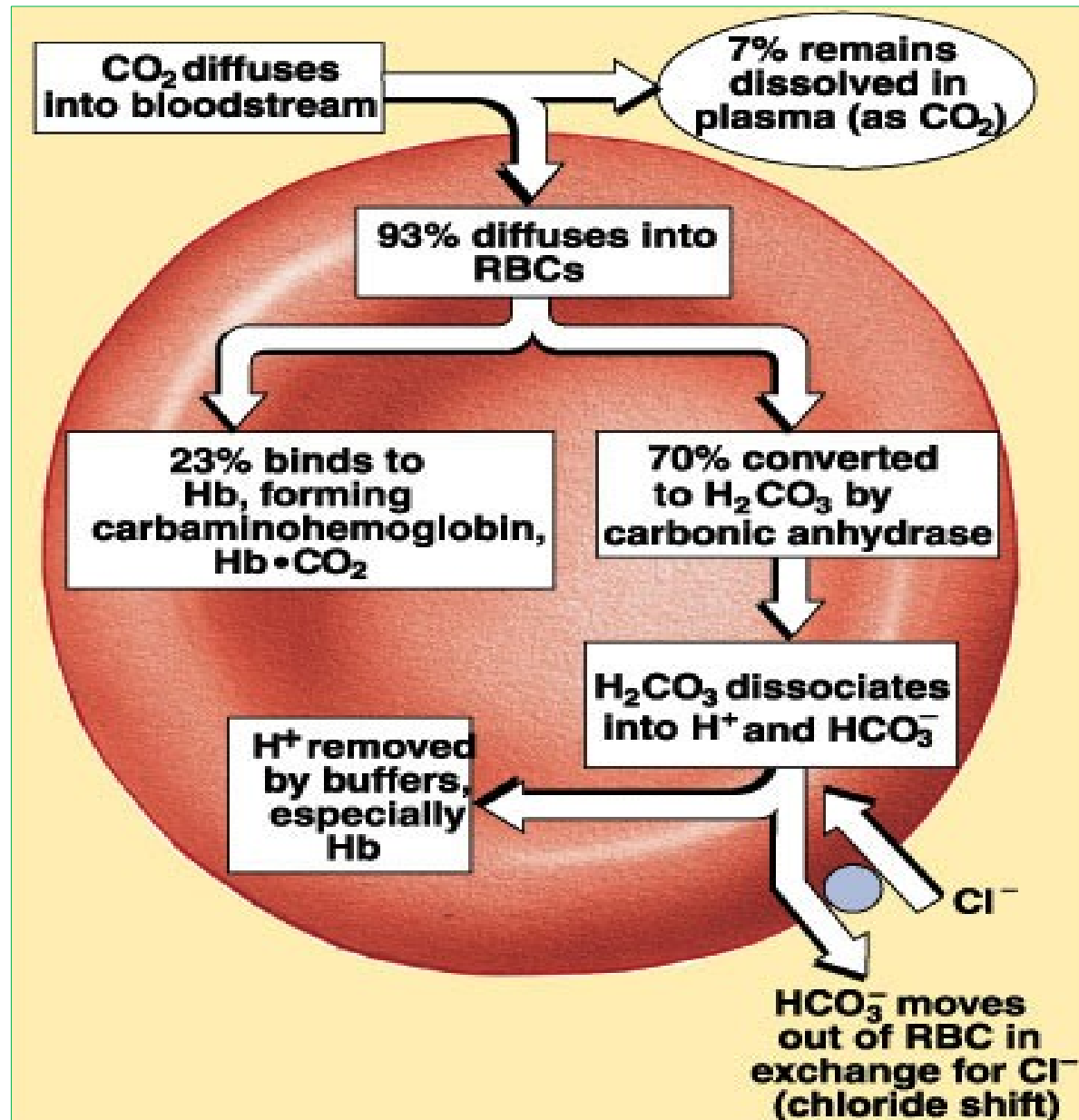
Transport of Carbon Dioxide...

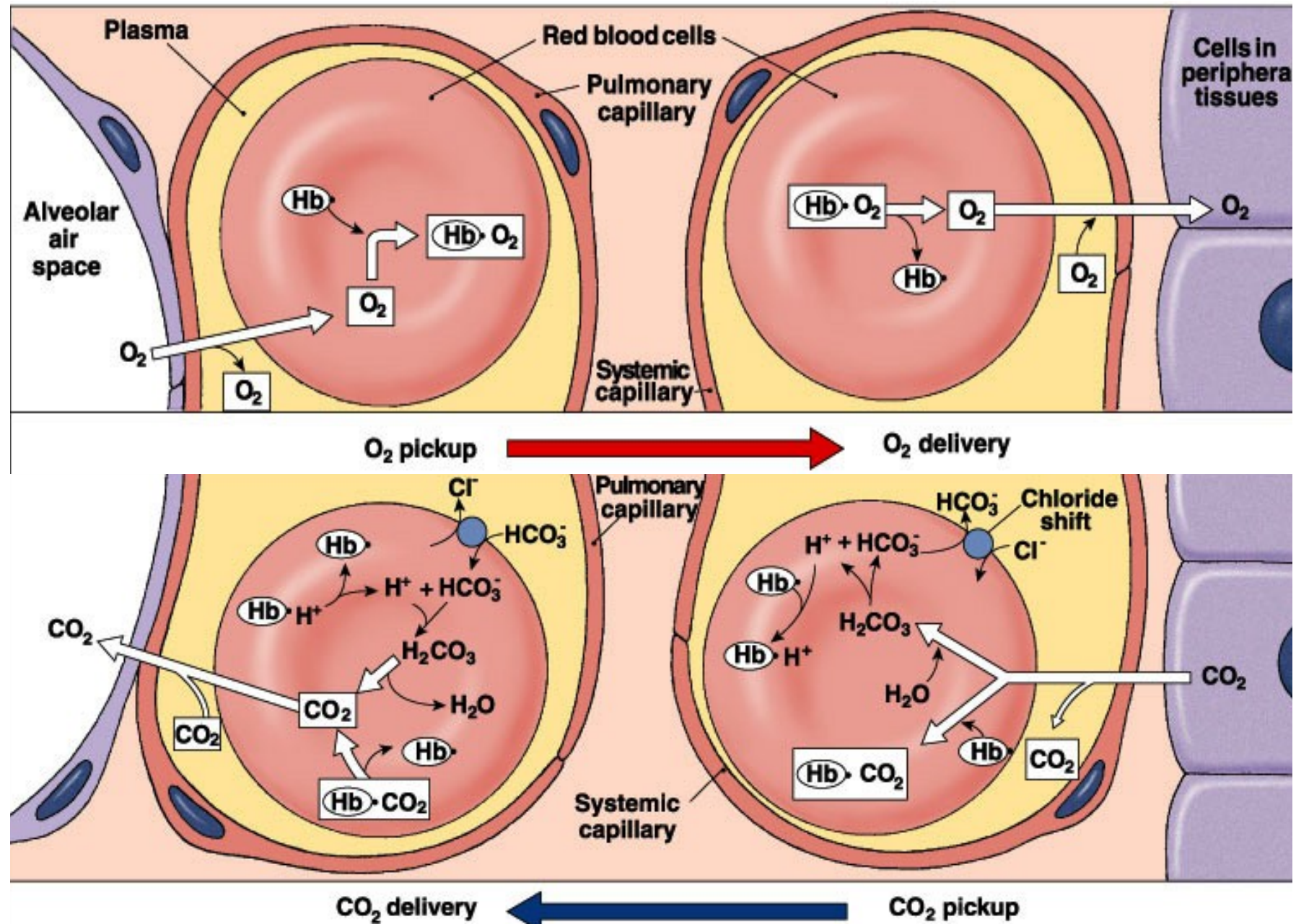
- **In tissue capillaries:**
 - **CO₂** combines with **water** inside **RBCs** to form **carbonic acid** which dissociates to form **bicarbonate ions** & **hydrogen ions**.
- **In lung capillaries:**
 - **bicarbonate ions** & **hydrogen ions** move into **RBCs** & **chloride ions move out**.

Transport of Carbon Dioxide...

- Bicarbonate ions combine with hydrogen ions to form **carbonic acid**, which is converted to carbon dioxide & water.
- The CO₂ diffuses out of the RBCs.
- ↑ in plasma CO₂ ↓ blood pH.
- Respiratory system regulates blood pH by regulating plasma carbon dioxide levels.

CO_2 transport & Cl^- movement





Pulmonary Circulation

- ❖ High-pressure, low-flow circulation: Bronchial vessels
 - ❑ Bronchial arteries
 - ❑ Bronchial veins

!!!!!! Bronchial veins to Pulmonary veins

Lymphatics???

Into the right thoracic lymph duct.

- ❖ Low-pressure, high-flow circulation: Pulmonary Vessels
 - ✓ Pulmonary arteries
 - ✓ Pulmonary veins

Regulation of Respiration

RESPIRATORY CENTER: medulla oblongata and pons of the brain stem

